Patient Characteristics and Individual Medical Costs Billed by Health Insurance

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Health insurers set premium and plan designs based on how much they expect to pay for each person they cover, so understanding which characteristics are associated with higher or lower medical costs is extremely important when determining insurance and health pricing. Characteristics such as age, sex, body mass index, smoking status, number of children and place of residence may all be related to how much a person pays for their health insurance.

The main research question of this analysis is: how do a person's age, sex, body mass index, number of dependents, smoking status, and region of residence relate to the individual medical costs billed by their health insurance?

1 Data Description and Processing

The data set used in this analysis was collected from health insurance companies about billed medical costs and the attributes of the primary beneficiary. We categorized the variables as personal variables and health-related variables:

Personal and Household Variables:
age: Age of the primary beneficiary in years. numeric
sex: Sex of the primary beneficiary categorical: female or male
children: Number of children (dependents) covered by the health insurance. numeric
region: The beneficiary's residential area in the US. categorical: northeast, southeast, southwest, or northwest

Health-Related Variables:
bmi: Body mass index of the beneficiary numeric
smoker: whether the beneficiary smokes. categorical: yes or no

Outcome:
charges: Individual medical costs billed by the health insurance in US dollars. numeric

The data set was checked for missing values. Of the original 1338 observations, we removed 5 observations that contained missing values leaving 1333 complete observations. We did not remove any outliers.

2 Data Exploration

The distribution of the outcome variable, medical charges, is shown in the figure below. It is strongly right-skewed: most beneficiaries have lower charges, with a long tail of much larger bills. There is also a second,

smaller bump at roughly $30,000 to $40,000.

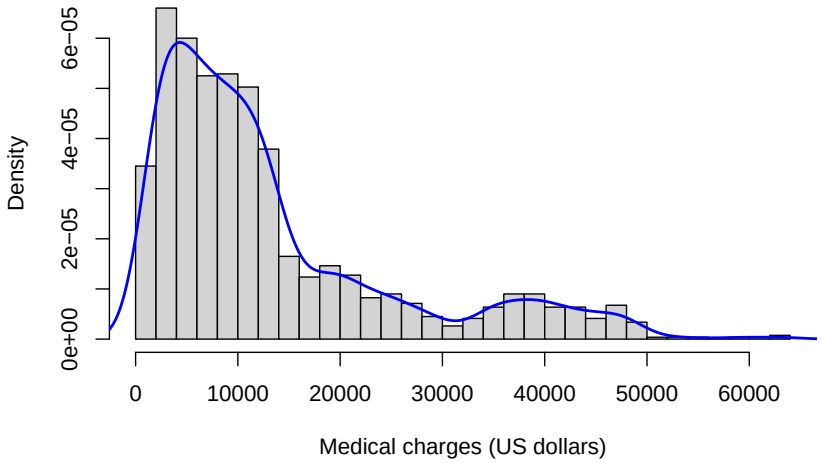


Figure 1: Distribution of individual medical charges.

The descriptive statistics are in the table below. The mean of $13282.32 is well above the median of $9386.16, which reflects the long right tail. Charges range from $1121.87 to $63770.43.

Table 1: Medical charges descriptive statistics (US dollars)

statistic	value
Min	1121.87
1st Quant. (25%)	4738.27
Mean	13282.32
Median	9386.16
3rd Quant. (75%)	16657.72
Max	63770.43
StdDev	12126.79

A skewed outcome is not automatically a problem for linear regression with a sample of this size. However, charges cannot fall below zero and the very large values can strongly influence a fitted model. Given these observations, we checked the residuals for problems in our fitted models.

2.1 Personal and Household Variables

We explored the distributions of the four personal and household variables.

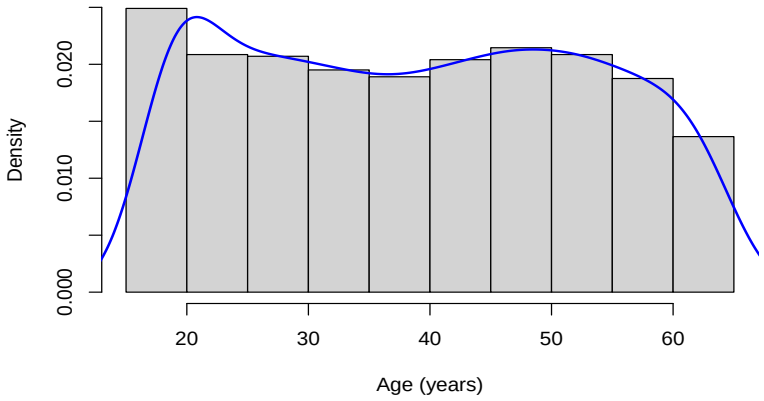


Figure 2: Distribution of age.

Table 2: Age descriptive statistics (years)

statistic	value
Min	18.00
1st Quant. (25%)	27.00
Mean	39.20
Median	39.00
3rd Quant. (75%)	51.00
Max	64.00
StdDev	14.06

Ages range from 18 to 64 and are spread fairly evenly across the range, with a mean and median close to 39. The mean and median are nearly the same, so the distribution is roughly symmetric.

We checked the frequencies of the two categorical variables, sex and region.

Table 3: Beneficiaries by sex

Sex	Count	Percent (%)
female	660	49.51
male	673	50.49

Table 4: Beneficiaries by region

Region	Count	Percent (%)
northeast	323	24.23
northwest	324	24.31
southeast	362	27.16
southwest	324	24.31

There are essentially equal numbers of males and females, and the four regions are well balanced, with the southeast slightly larger than the others. This means we have a good representation of each level.

The number of children is recorded as whole numbers, so we plotted the number of beneficiaries at each value.

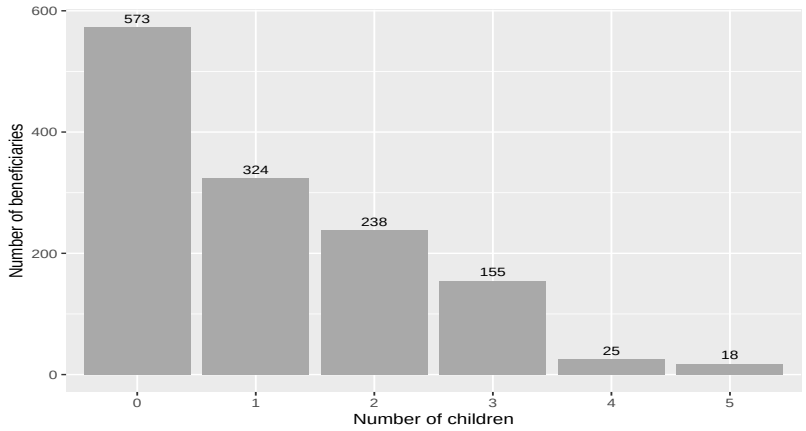


Figure 3: Distribution of the number of children.

The figure above shows that the most common number of children is 0, and the counts fall steadily as the number of children increases. Having four or five children is rarer.

2.2 Health-Related Variables

We explored the two health-related variables. Body mass index is shown in the figure below.

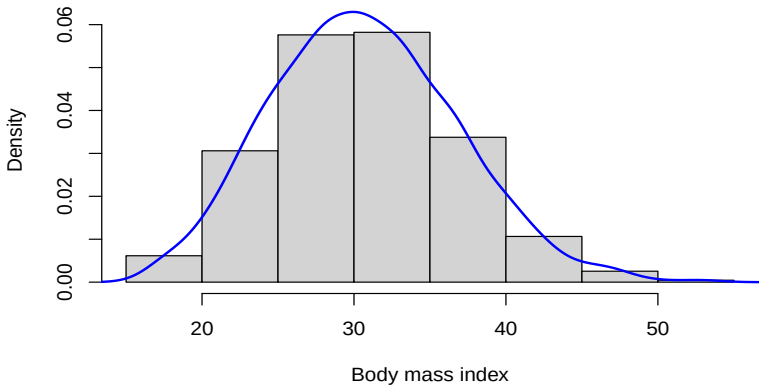


Figure 4: Distribution of body mass index.

Table 5: Body mass index descriptive statistics

statistic	value
Min	15.96
1st Quant. (25%)	26.32
Mean	30.68
Median	30.40
3rd Quant. (75%)	34.70
Max	53.13
StdDev	6.09

Body mass index has a roughly bell-shaped distribution with a slight tail toward higher values. Most beneficiaries fall between about 25 and 35, and the mean and median are both close to 30.

Smoking status is the second health-related variable.

Table 6: Beneficiaries by smoking status

Smoker	Count	Percent (%)
no	1060	79.52
yes	273	20.48

Table 6 shows smokers are a much smaller group than non-smokers, at about 20.48% of beneficiaries, so estimates for smokers rest on fewer observations.

3 Modeling

To answer our research question, we fitted a multiple linear regression model to see how a person's characteristics relate to the medical costs billed by their health insurance.

3.1 Model 1: All Predictors

The first model uses all six predictors: age, sex, bmi, children, smoker and region. The dependent variable is charges and the other six variables are the independent variables. The categorical predictors are included as dummy variables, with one level of each left out as the baseline: female for sex, no for smoker, and northeast for region. The estimates for the coefficients, standard errors and p-values are shown in table 7.

Table 7: Summary of coefficients for model 1.

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	-11887.48	990.06	-12.01	0.0000
age	257.02	11.93	21.55	0.0000
sexmale	-119.43	333.81	-0.36	0.7206
bmi	336.54	28.75	11.71	0.0000
children	479.19	138.28	3.47	0.0005
smokeryes	23885.10	414.07	57.68	0.0000
regionnorthwest	-343.85	477.32	-0.72	0.4714
regionsoutheast	-996.88	480.45	-2.07	0.0382
regionsouthwest	-951.83	479.02	-1.99	0.0471

The adjusted R-squared for this model was 0.75, so the model explains about 75% of the variance in medical charges. The F-statistic was 499.97 on 8 and 1324 degrees of freedom with a p-value less than 0.0001, which means the model is statistically significant overall.

Age, bmi, children and smoker are all highly significant with p-values < 0.001. Two of the three region levels, southeast and southwest, are also significant ($p < 0.05$), but northwest is not. Sex is the only predictor that is not significant ($p = 0.72$), so there is no real difference in charges between males and females once the other predictors are held constant. Table 8 ranks the significant predictors by the size of their coefficients.

Table 8: Ranking of predictors' importance in model 1.

Rank	Predictor	Category	Coefficient
1	smokeryes	health-related	23885.10
2	regionsoutheast	personal and household	-996.88
3	regionsouthwest	personal and household	-951.83
4	children	personal and household	479.19
5	bmi	health-related	336.54
6	age	personal and household	257.02

Smoking has by far the biggest effect on medical charges. On average, smokers are billed about $23,885.10 more than non-smokers, holding the other predictors constant. That is more than 20 times bigger than any other coefficient in the model. The two significant region levels come next: people in the southeast and southwest are billed about $996.88 and $951.83 less than people in the northeast, but these are also the least significant predictors in the table ($p = 0.0382$ and $p = 0.0471$). Each extra child on the plan adds about $479.19 in charges, each 1 unit increase in body mass index adds about $336.54, and each extra year of age adds about $257.02.

3.1.1 Evaluation of Model 1 Fit

We calculated the variance inflation factor (VIF) for each predictor to check for multicollinearity. Since region is categorical with more than two levels, the table shows the generalized VIF (GVIF), along with the adjusted GVIF that is used to compare categorical predictors.

Table 9: VIF calculations for model 1.

	GVIF	Df	GVIF^(1/(2*Df))
age	1.02	1	1.01
sex	1.01	1	1.00
bmi	1.11	1	1.05
children	1.00	1	1.00
smoker	1.01	1	1.01
region	1.10	3	1.02

All of the values in table 9 are close to 1, so there is no multicollinearity in the model and we did not need to remove any predictors.

The residuals vs. fitted values plot and the Q-Q plot for model 1 are shown in figure 5.

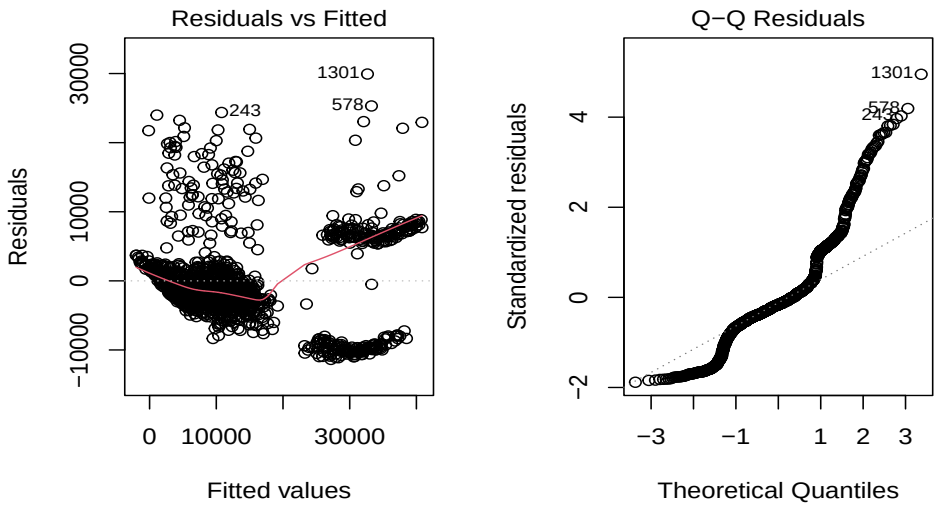


Figure 5: Plot of residuals vs. fitted values and Q-Q plot for model 1.

The residuals are not randomly scattered around 0. For lower fitted values, most of the points are bunched up just under the zero line, with a lot of points far above it. For higher fitted values, the residuals split into two separate groups, one above the zero line and one below it. Patterns like this suggest the model is missing something, such as nonlinear terms or interactions between the predictors. The Q-Q plot shows the residuals are not normally distributed, with the points moving far away from the line at both ends, especially at the top. This matches the right-skewed distribution of charges in figure 1.

We also ran a Breusch-Pagan test to check for non-constant variance (heteroscedasticity).

Table 10: Breusch-Pagan test results.

BP statistic	df	p-value
121.06	8	0

The p-value in table 10 rounds to 0 (it is less than 0.0001), so we reject the null hypothesis and conclude that heteroscedasticity is present. Since this can make the standard errors and p-values unreliable, we recalculated them using robust (HC3) standard errors. The results are in table 11.

Table 11: Summary of coefficients for heteroskedasticity-robust (HC3) standard errors calculations for model 1.

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	-11887.48	1052.71	-11.29	0.0000
age	257.02	12.00	21.42	0.0000
sexmale	-119.43	336.05	-0.36	0.7223
bmi	336.54	32.00	10.52	0.0000
children	479.19	131.67	3.64	0.0003
smokeryes	23885.10	579.39	41.22	0.0000
regionnorthwest	-343.85	488.34	-0.70	0.4815
regionsoutheast	-996.88	504.95	-1.97	0.0486
regionsouthwest	-951.83	464.77	-2.05	0.0408

The coefficients stay the same, but some of the standard errors changed, especially for smoker (from 414.07 to 579.39). Even so, the same predictors are still significant at the 0.05 level as in table 7, so the heteroscedasticity does not change which predictors are related to medical charges.

To sum up the fit, model 1 explains a good amount of the variance in charges and smoking stands out as the most important predictor. There is no multicollinearity, and the robust standard errors show the significant predictors hold up even with the non-constant variance. Our biggest concern is the pattern in the residuals, which shows that this model doesn't fully capture how charges work, especially for people with high charges. We will keep this in mind when we choose the predictors for our second model.

4 Predicting with Model 1

To see how well model 1 predicts charges for data it hasn't seen, we used 10-fold cross validation. On each fold, 80% of the data was used to train the model and the other 20% was used to test it, with the data for each fold picked at random. We repeated the 10-fold cross validation 3 times, for a total of 30 folds, to get a better measure of how the model performs. We then calculated the average root mean squared error (RMSE) over the 30 folds and the standard deviation of the RMSE values, shown in table 12.

Table 12: Mean RMSE and standard deviation for model 1.

Model	Avg RMSE	Std Dev
Model 1	6080.86	335.88

On average, the model's predictions were off by about \$6,080.86. That is about 46% of the mean charge of \$13,282.32, which is a pretty large error. This makes sense given the problems we saw in the residuals. We plotted the RMSE for each of the 3x10=30 folds in figure 6.

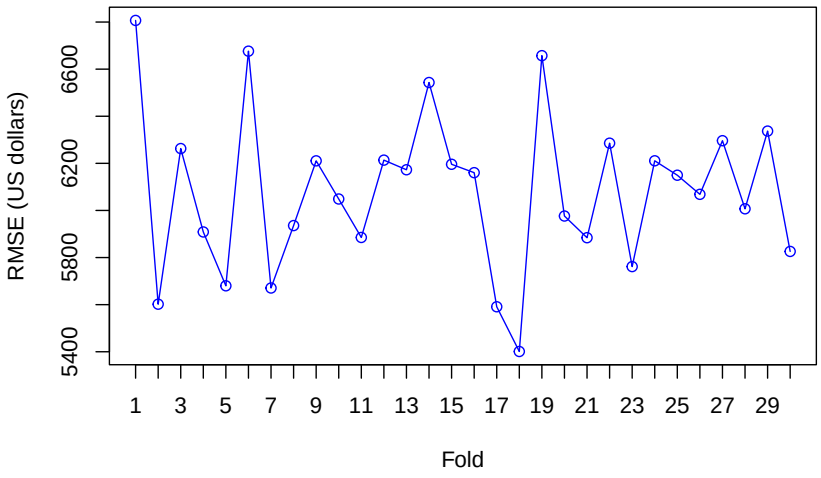


Figure 6: RMSE over 3x10=30 folds for model 1.

Figure 6 shows the RMSE stayed in a fairly small range over all 30 folds, from about \$5,401 to about \$6,807. The standard deviation is small compared to the average RMSE, so the model's error was about the same no matter which part of the data it was trained and tested on. So model 1 is consistent in its predictions, but they are not very precise.